Tom 2 15/02/23

Reg. No. | | | | | | | | | | | [22BSPH11]

FIRST SEMESTER B. Tech EXAMINATION, FEBRUARY - 2023

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

Physical Constants :

i. Mass of the electron $m_e = 9.1 \times 10^{-31} kg$

ii. Mass of proton, neutron $m_n, m_p = 1.67 \times 10^{-27} kg$

iii. Charge of the electron $e = 1.602 \times 10^{-19} C$

iv. Planck's constant $h = 6.626 \times 10^{-34} Js$

v. Permittivity of free space $\varepsilon_o = 8.85 \times 10^{-12} F/m$

vi. Velocity of light $c = 3 \times 10^8 m/s$

vii. Permeability of free space $\mu_o = 4\pi \times 10^{-7} H/m$

ANSWER ALL QUESTIONS

		PART - A			**5 X 2=10**
1.	**a.**	Define phase velocity and group velocity.	L-1	CO1	**02**
	b.	Explain the conditions for LASER action.	L-2	CO2	**02**
	c.	Define Fermi Energy and Fermi Velocity.	L-1	CO3	**02**
	d.	Outline the properties of shock waves.	L-2	CO4	**02**
	e.	Define Young's modulus and Rigidity modulus.	L-1	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Describe de-Broglie's hypothesis of matter waves.	L-2	CO1	**05**
		OR			
	b.	State and explain Heisenberg's uncertainty principle.	L-2	CO1	**05**
3.	**a.**	What is population inversion? Describe how the population inversion can be achieved using the metastable state.	L-2	CO2	**05**
		OR			
	b.	Deduce an expression for Numerical aperture in terms of refractive indices of core and cladding of an optical fiber.	L-3	CO2	**05**
4.	**a.**	Discuss the aspects in which the classical free electron theory has failed.	L-2	CO3	**05**
		OR			
	b.	Outline the assumptions of quantum free electron theory for conductors.	L-2	CO3	**05**
5.	**a.**	**i.** Differentiate between intrinsic and an extrinsic semiconductor. **ii.** Explain the effective mass of an electron.	L-4 L-2	CO3 CO3	**03** **02**
		OR			

	b.	With the help of a neat diagram, discuss how the photodiode conducts in reverse bias.	L-2	CO3	**05**
6.	a.	For simple harmonic motion, define the following **i.** Amplitude **ii.** Displacement **iii.** Acceleration **iv.** Frequency **v.** Phase	L-1	CO4	**05**
		OR			
	b.	Obtain an expression for force constant for springs which are parallelly connected to each other.	L-2	CO4	**05**
7.	a.	Mention the types of beams and its applications.	L-4	CO6	**05**
		OR			
	b.	Define Stress and Strain. Paraphrase the types of elastic moduli.	L-2	CO6	**05**
		PART - C		**6 X 10=60**	
8.	a.	**i.** Describe Davison and Germer's experiment to show that the electron behaves as a wave. **ii.** Calculate the de-Broglie wavelength and group velocity of the de-Broglie wave of an electron having a kinetic energy of 100 eV.	L-2 L-3	CO1 CO1	**07** **03**
		OR			
	b.	Using one dimensional time independent Schrodinger's wave equation to particle in a potential box of width "a" to find out the energy eigen values and obtain the normalized wave function.	L-3	CO1	**10**
9.	a.	**i.** Deduce an expression for energy density in terms of Einstein's A & B coefficients. **ii.** Find the number of photons emitted per second by a He-Ne laser source emitting light of wavelength 632.8 nm with an optical power of 10mW.	L-2 L-3	CO2 CO2	**07** **03**
		OR			
	b.	**i.** Discuss induced absorption, spontaneous emission and stimulated emission. **ii.** With the help of a neat diagram, explain the types of optical fibers.	L-2 L-2	CO2 CO2	**04** **06**
10.	a.	**i.** Explain how the quantum free electron theory successfully overcome the failures of classical free electron theory. **ii.** Find the temperature at which there is 1% probability that a state with an energy 0.5eV above Fermi energy is occupied.	L-2 L-3	CO3 CO3	**06** **04**
		OR			

	b.	i. Explain the variation of Fermi factor with temperature and energy. ii. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	L-2 L-3	CO3 CO3	**06** **04**
11.	a.	i. Using the expression for charge career concentration for electrons and holes, show that Fermi level lies exactly in between the valence band and conduction band in case of intrinsic semiconductor. ii. Use the following given data for intrinsic germanium at 330 K to find the resistivity. $n_i = 2.4 \times 10^{19}/m^3$, $\mu_e = 0.39\ m^2v^{-1}s^{-1}$, $\mu_h = 0.19\ m^2v^{-1}s^{-1}$.	L-2 L-3	CO4 CO4	**06** **04**
		OR			
	b.	Discuss the procedure to find the band gap of a semiconductor experimentally with necessary equations.	L-2	CO4	**10**
12.	a.	Deduce an expression for the decay of the amplitude in damped oscillations.	L-2	CO5	**10**
		OR			
	b.	Deduce an expression for amplitude and phase of forced vibrations.	L-2	CO5	**10**
13.	a.	i. Arrive at an expression for Young's modulus using single cantilever. ii. Calculate the force required to produce an extension of 1 mm in steel wire of length 2 meters and diameter 1 mm. (Young's modulus for steel $Y = 2 \times 10^{11} Nm^{-2}$)	L-3 L-3	CO6 CO6	**07** **03**
		OR			
	b.	Obtain the relation between Young's modulus (Y), Rigidity modulus(η) and Poisson's Ratio (σ)	L-2	CO6	**10**